

# Karthik Viswanathan

✉ vkarthik095@gmail.com • 🏠 Website • 🎓 Scholar • 🔗 LinkedIn • 🐙 GitHub

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## Education

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### University of Amsterdam

*PhD in Physics, Defending in March 2026*

**Thesis:** From Language Models to Cosmic Structures: A Geometric Perspective

**Advisor:** Dr. Jan Pieter van der Schaar

**Amsterdam**

2021 – 2025

### University of Amsterdam

*MSc in Physics and Astronomy (Theoretical Physics), cum laude*

**Thesis:** Exploring the Spectral Theory/Topological Strings duality

**Advisor:** Dr. Marcel Vonk

**GPA:** 8.7/10

**Amsterdam**

2019 – 2021

### Indian Institute of Technology Madras

*B.Tech in Engineering Physics*

**Thesis:** Real Space Renormalization and Applications to Machine Learning

**Advisor:** Dr. Ashwin Joy

**GPA:** 8.29/10

**Chennai**

2013 – 2017

## Publications

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### Probing Geometry of Next Token Prediction Using Cumulant Expansion of Softmax Entropy

Karthik Viswanathan, Sang Eon Park

ICML 2025: Workshop on High-dimensional Learning Dynamics [[paper](#), [code](#)]

### Persistent Topological Features in Large Language Models

Yuri Gardinazzi\*, Karthik Viswanathan\*, Giada Panerai, Alessio Ansuini, Alberto Cazzaniga, Matteo Biagetti

ICML 2025 (Poster) [[paper](#), [code](#)]

### The Geometry of Tokens in Internal Representations of Large Language Models

Karthik Viswanathan, Yuri Gardinazzi, Giada Panerai, Alberto Cazzaniga, Matteo Biagetti

Under review [[paper](#), [code](#)]

### Cosmology with persistent homology: A Fisher forecast

Jacky Yip, Matteo Biagetti, Alex Cole, Karthik Viswanathan, Gary Shiu

Journal of Cosmology and Astroparticle Physics (JCAP), Vol. 2024, Issue 09 [[paper](#), [code](#)]

## Research Programs & Workshops

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### ML Alignment & Theory Scholars Program (MATS 9.0)

Mentors: Adam Shai & Paul Riechers (Simplex AI Safety)

**Berkeley**

Jan – Mar 2026

### Mentorship for Alignment Research Students (MARS 3.0)

Mentors: Puria Radmard, Cameron Tice & Edward Young (Geodesic Research)

Architectures for Increased Externalisation of Reasoning [[blogpost](#), [code](#)]

**Cambridge**

Jul – Sep 2025

## Workshop: Interpretability in LLMs using Geometric and Statistical methods

Organized by Karthik Viswanathan, Jan Pieter van der Schaar

Amsterdam

May 2025

## Professional Experience

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### Goldman Sachs

*Surveillance Analyst*

Developed ML models for anomaly detection in financial data

**Manager:** Prof. Howard Karloff

Bangalore

2017 – 2019

### Goldman Sachs

*Summer Intern*

Implemented fast Personalized PageRank in a MapReduce framework

**Mentor:** Dr. Koushik Balasubramanian

Bangalore

2016

## Awards & Distinctions

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### Sander Bais Prize for Academic Merit

*Institute for Theoretical Physics Amsterdam*

For exceptional academic performance in the master's program

Amsterdam

2020

### Honorable Mention

*ACM ICPC World Finals*

Represented India in the international collegiate programming contest.

Marrakech

2015

### Participant

*International Olympiad in Informatics Training Camp*

Selected after ranking in top 25 in the Indian National Olympiad in Informatics

Bangalore

2013

## Organization & Mentoring

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### AREA Science Park, Trieste

*Master Student Supervisor*

**Student:** Giada Panerai

**Thesis:** Zig-Zag Persistence in Neural Networks Representations

2024

### University of Amsterdam

*Master Student Mentor*

**Student:** Sibilla Bouche

**Thesis:** The persistence of non-Gaussian features: a Neural Ratio Estimation approach

2023

### University of Amsterdam

*Master Student Supervisor*

**Student:** Sliem el Ela

**Thesis:** From Primordials to Persistence: A Dual Exploration of Multiparameter Topology and Cosmic Origins

2023

### University of Amsterdam

*Journal Club Organizer*

**Cosmology Journal Club**

2022

## Research Visits

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2024: AREA Science Park, Trieste, Italy

September 16 – October 4

2024: AREA Science Park, Trieste, Italy

February 15 – April 27

## Technical Skills

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**Programming:** PyTorch, TensorFlow, Hadoop, MapReduce, Mathematica

**HPC Resources:** LUMI consortium, Snellius supercomputer

## Teaching

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**University of Amsterdam**

**MSc Physics Program**

*Teaching Assistant*

2020 – 2024

**Courses:** Topological Data Analysis: a Physics Perspective, Advanced Cosmology, ML for Physics and Astronomy, QFT in Curved Spacetimes, General Relativity

## References

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**Dr. Jan Pieter van der Schaar**

University of Amsterdam

j.p.vanderschaar@uva.nl

**Dr. Magnus Botnan**

Vrije Universiteit, Amsterdam

M.B.Botnan@vu.nl

**Dr. Matteo Biagetti**

AREA Science Park, Trieste

matteo.biagetti@areasciencepark.it

**Dr. Alberto Cazzaniga**

AREA Science Park, Trieste

alberto.cazzaniga@areasciencepark.it